A brief introduction to:

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Statistics in Space

Building the intuition for spatial and spatiotemporal generalized linear mixed models (GLMMs)

Generalized linear models and maximum likelihood

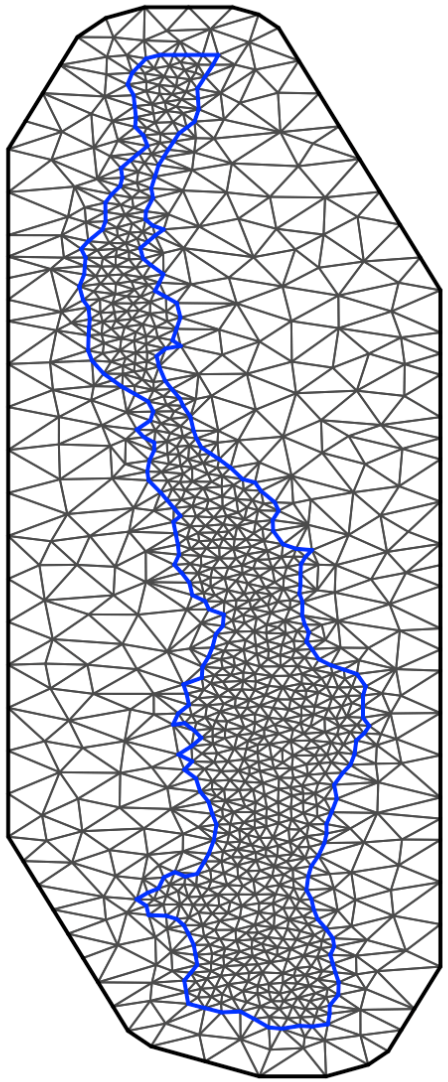
- Simulated quantile-based residuals and evaluating residual spatial autocorrelation

GLMMs and the Laplace approximation

Spatial and spatiotemporal GLMMs

- Gaussian Fields, Matérn covariance
- The SPDE method

Follow along in R: `R/statistics_in_space_intro.r`



A triangulated Chesapeake Bay

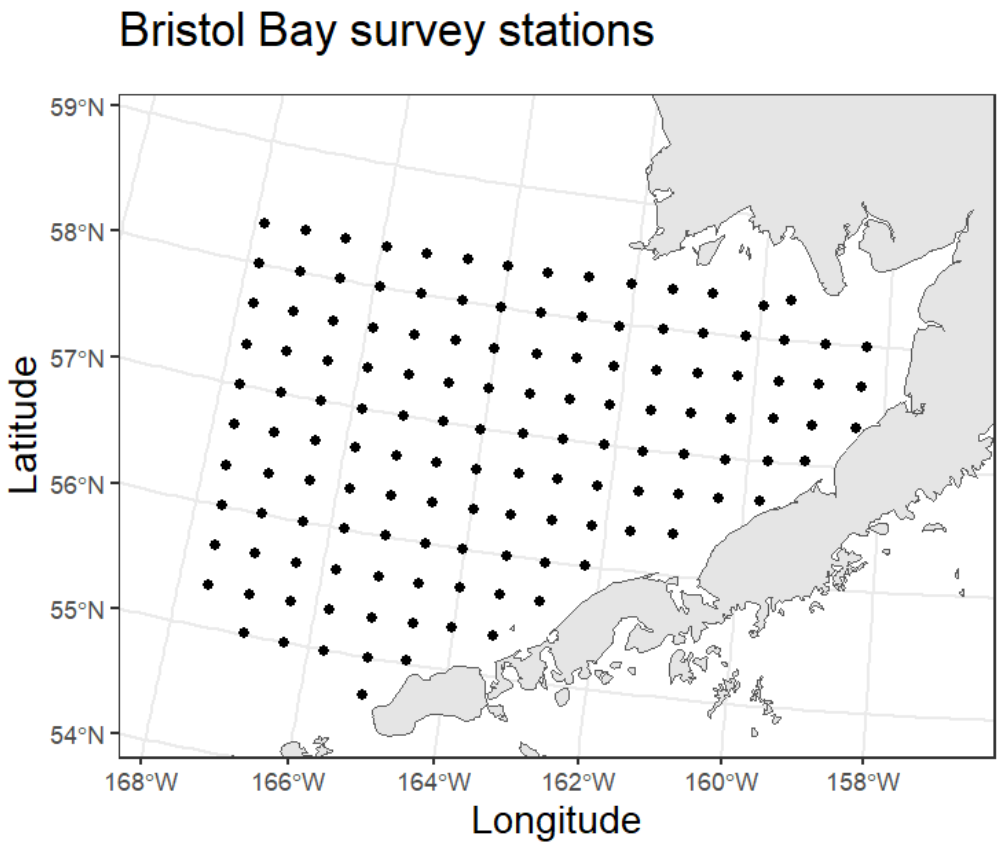
Generalized linear models

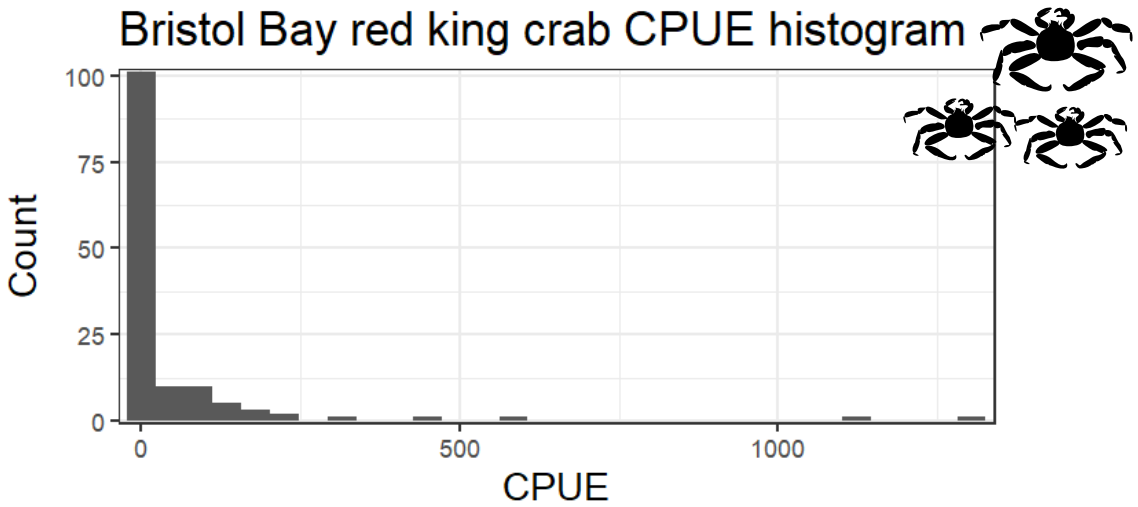
GLMs allow for modeling observations drawn from diverse *probability distributions*

- Gaussian, Poisson, negative binomial, Tweedie etc

Question: What was the mean catch per unit effort (CPUE) of red king crab in Bristol Bay in the 2010 NMFS bottom trawl survey?

- CPUE must be ≥ 0 and commonly right-skewed
- Tweedie probability distribution could be good *data model* for CPUE
  - Useful for continuous positive data with zeros present (e.g., biomass/numbers density)





Generalized linear models

Approach: Fit an intercept-only, Tweedie GLM to the data

The model requires specifying:

1. A **data model**

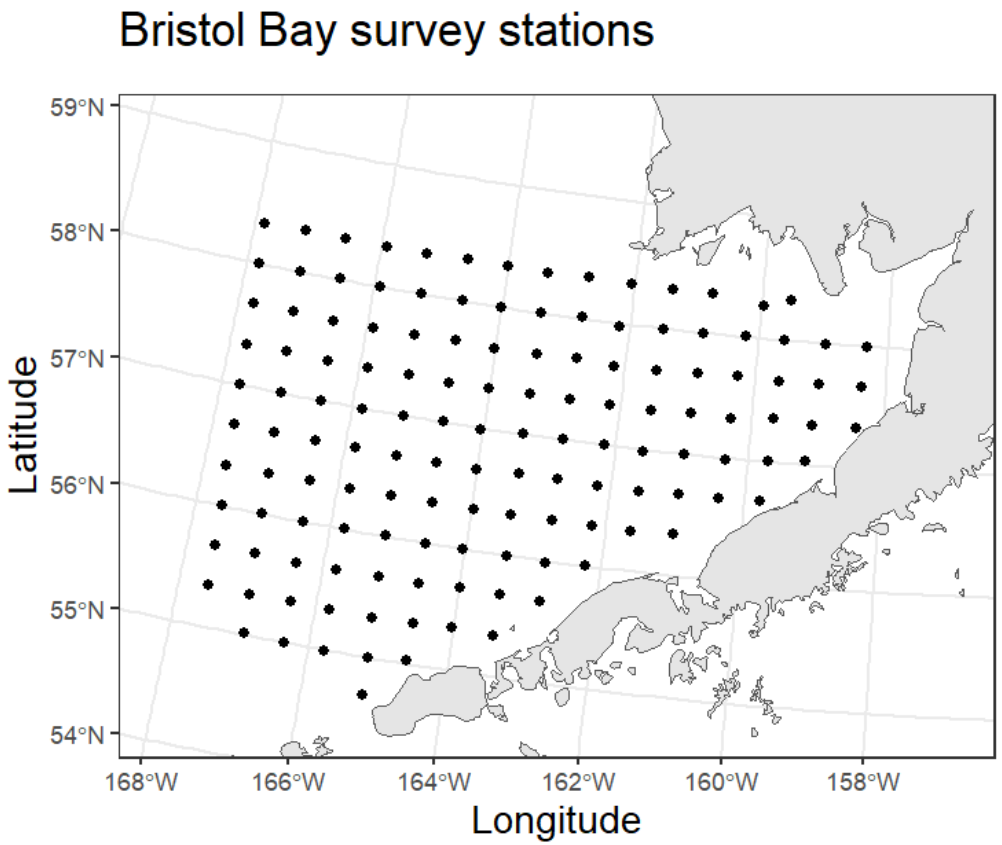
$$\mathrm{CPUE}_s \sim \mathrm{Tweedie}(\mu_s, \phi, p)$$

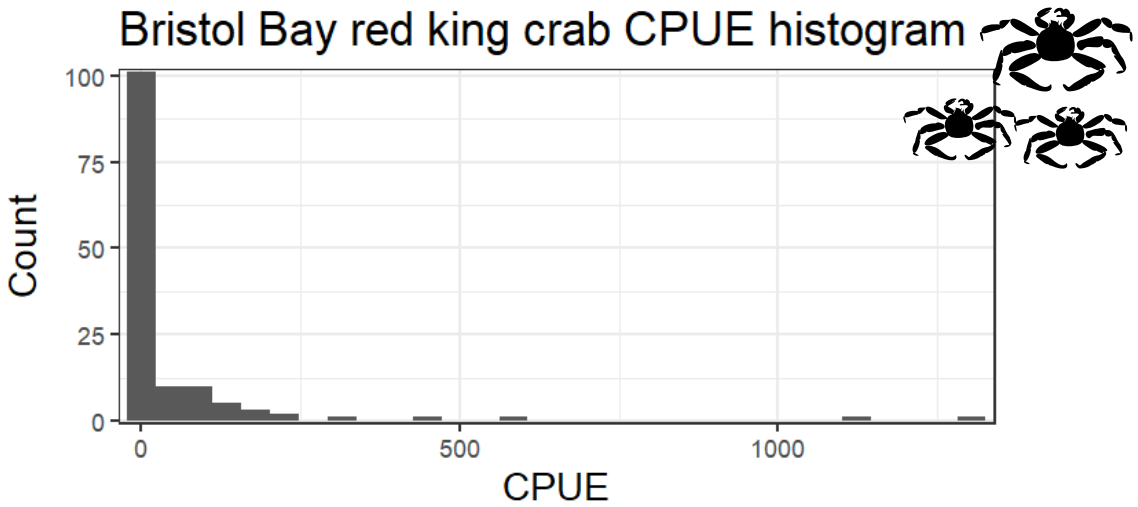
2. A **linear predictor** with link function

$$\log(\mu_s) = \beta_0$$

β_0, ϕ, p are parameters that need to be estimated

- β_0 is log-CPUE at sampling location *s*
- ϕ and p are parameters characterizing Tweedie shape



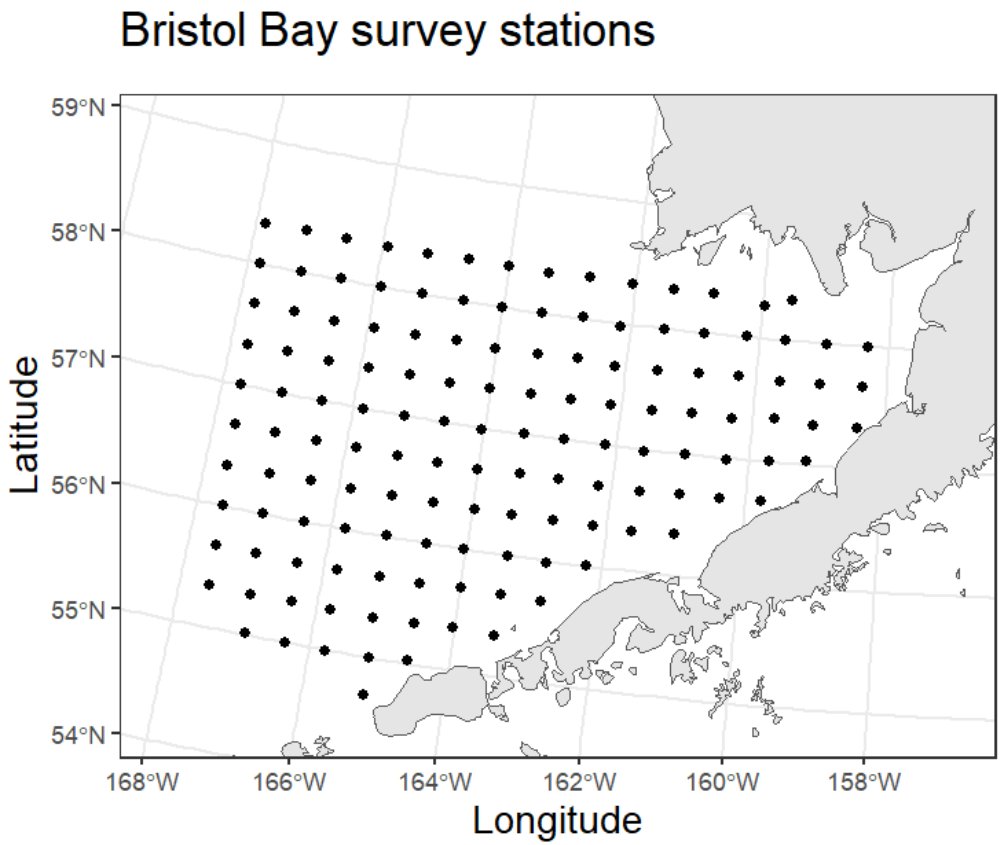


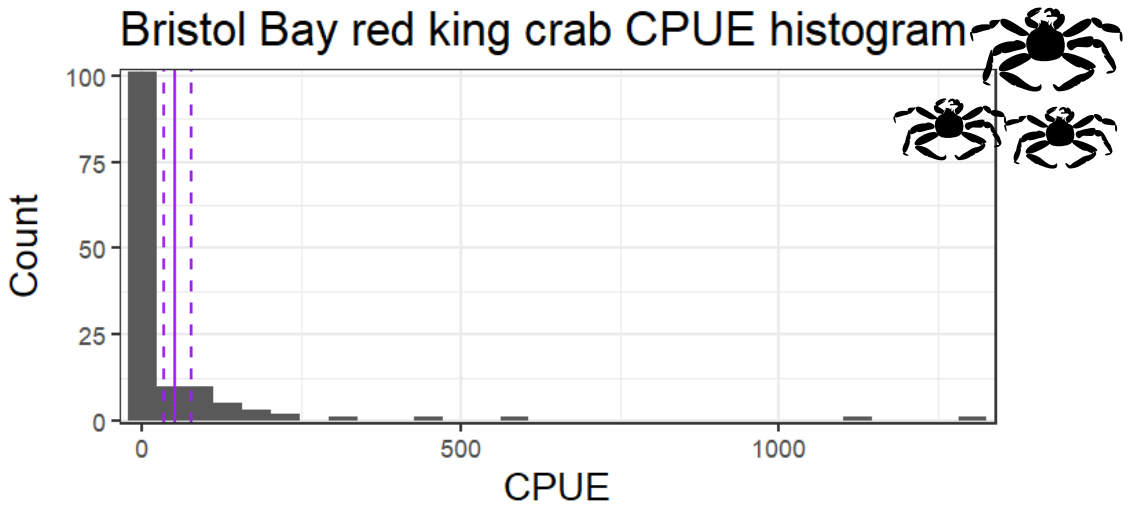
Generalized linear models

Fitting the model in *sdmTMB* via maximum likelihood estimation (MLE)

```
> # intercept only GLM-----------------------------------
> mod1 <- sdmTMB(cpue ~ 1,
+                family = sdmTMB::tweedie(),
+                spatial = "off",
+                data = rkc_cpue_2010_df)
> summary(mod1)
Model fit by ML ['sdmTMB']
Formula: cpue ~ 1
Mesh: NULL (isotropic covariance)
Data: rkc_cpue_2010_df
Family: tweedie(link = 'log')

Conditional model:
            coef.est coef.se
(Intercept)     3.91    0.21

Dispersion parameter: 39.57
Tweedie p: 1.52
ML criterion at convergence: 328.084

See ?tidy.sdmTMB to extract these values as a data frame.
```

The answer to our question is $\mu_s = exp(\beta_0) = 50.12$ crabs per sq. km





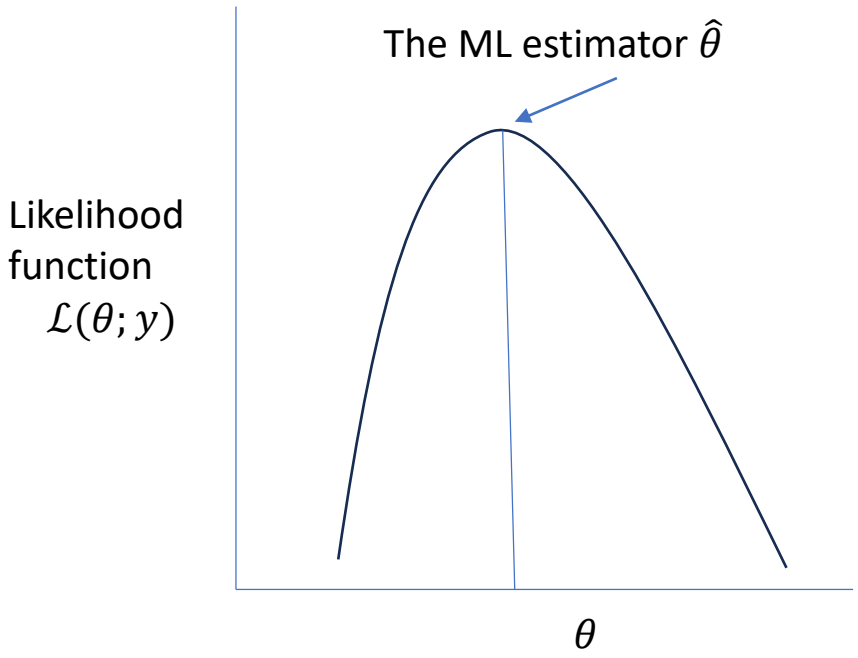
Spare a moment for maximum likelihood estimation

MLE for GLM involves:

1. Specifying a probability model for the data (Tweedie here)

2. Using the model to calculate the likelihood $\mathcal{L}(\theta; y)$ of observing each observation y given a set of parameters θ

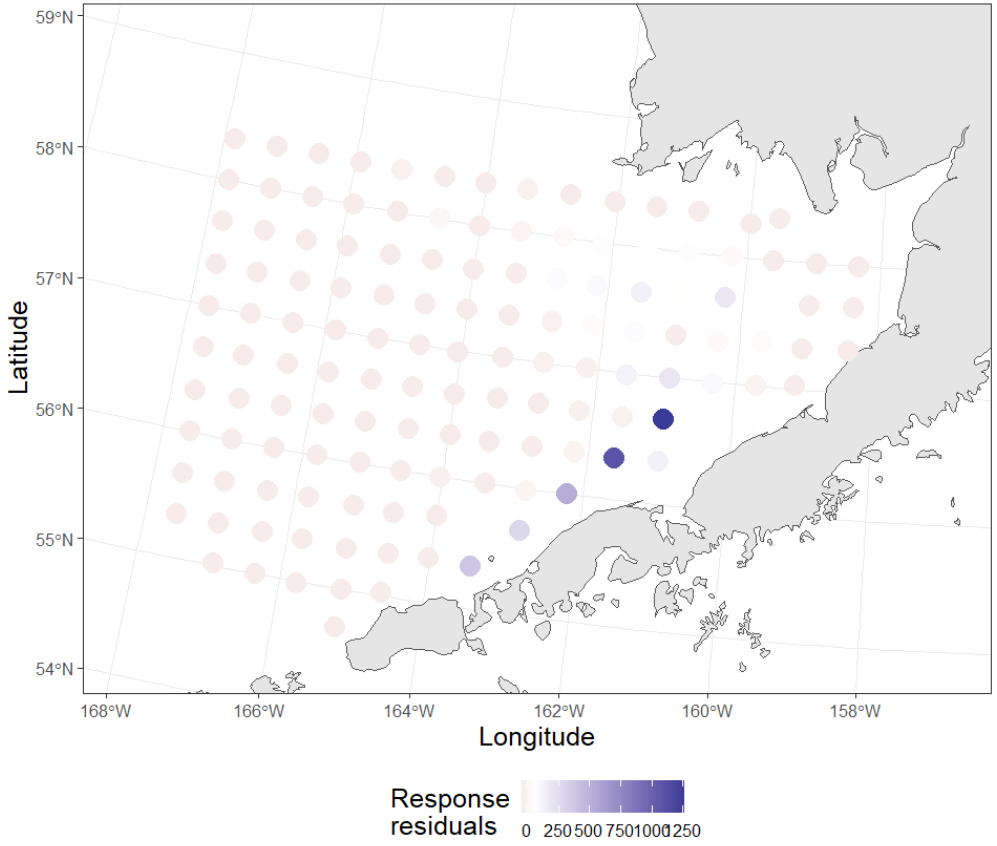
3. Identifying the parameters $\hat{\theta}$ that maximize the joint likelihood of all observations using an optimization algorithm
   - In practice, we minimize the joint negative log-likelihood of the data



MLE assumes residual independence

- **Violations of this assumption can bias inference and uncertainty**

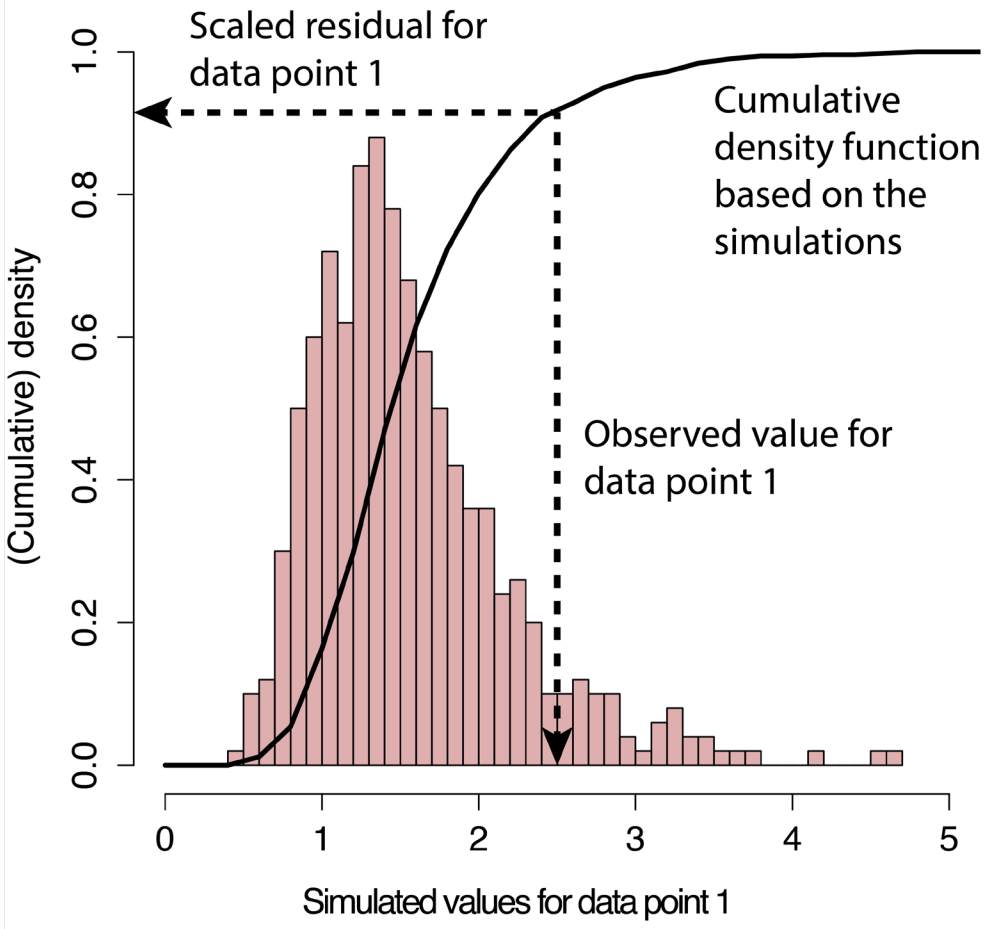
Are residuals independent?



- We expect that they are not, but for non-Gaussian GLMs evaluating residual patterns on the response scale is often unhelpful

- Instead, we introduce **simulation-based quantile residuals** aka **DHARMa residuals**

DHARMa residuals



Simulation-based quantile residuals:

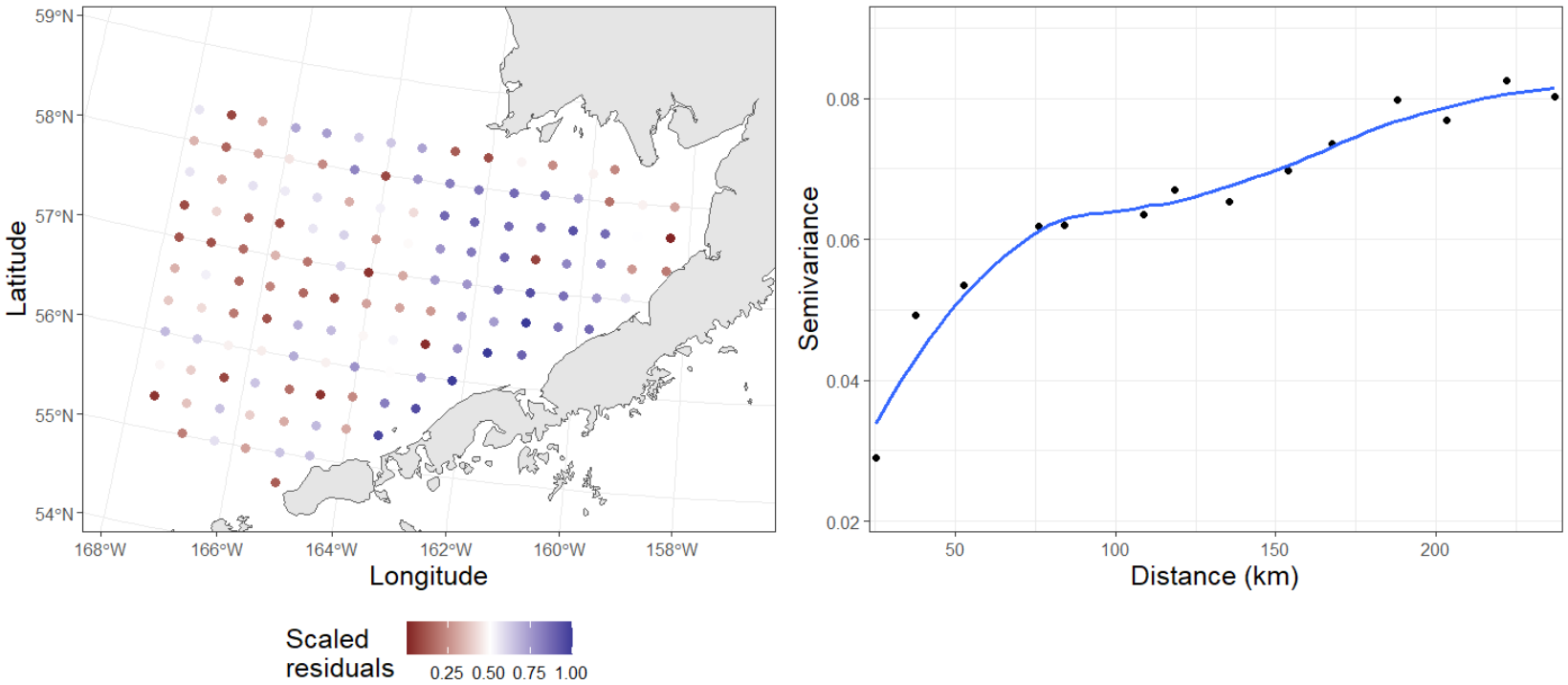
1. We simulate observations from the fitted model to provide the expected distribution for each observation.

2. The fraction of simulated observations $\leq$ the observation is the **quantile residual**

Well-behaved DHARMa residuals will be uniformly distributed and are returned as “scaled residuals”

Works for diverse response families and random effect structures

Back to crabs. Are residuals independent?

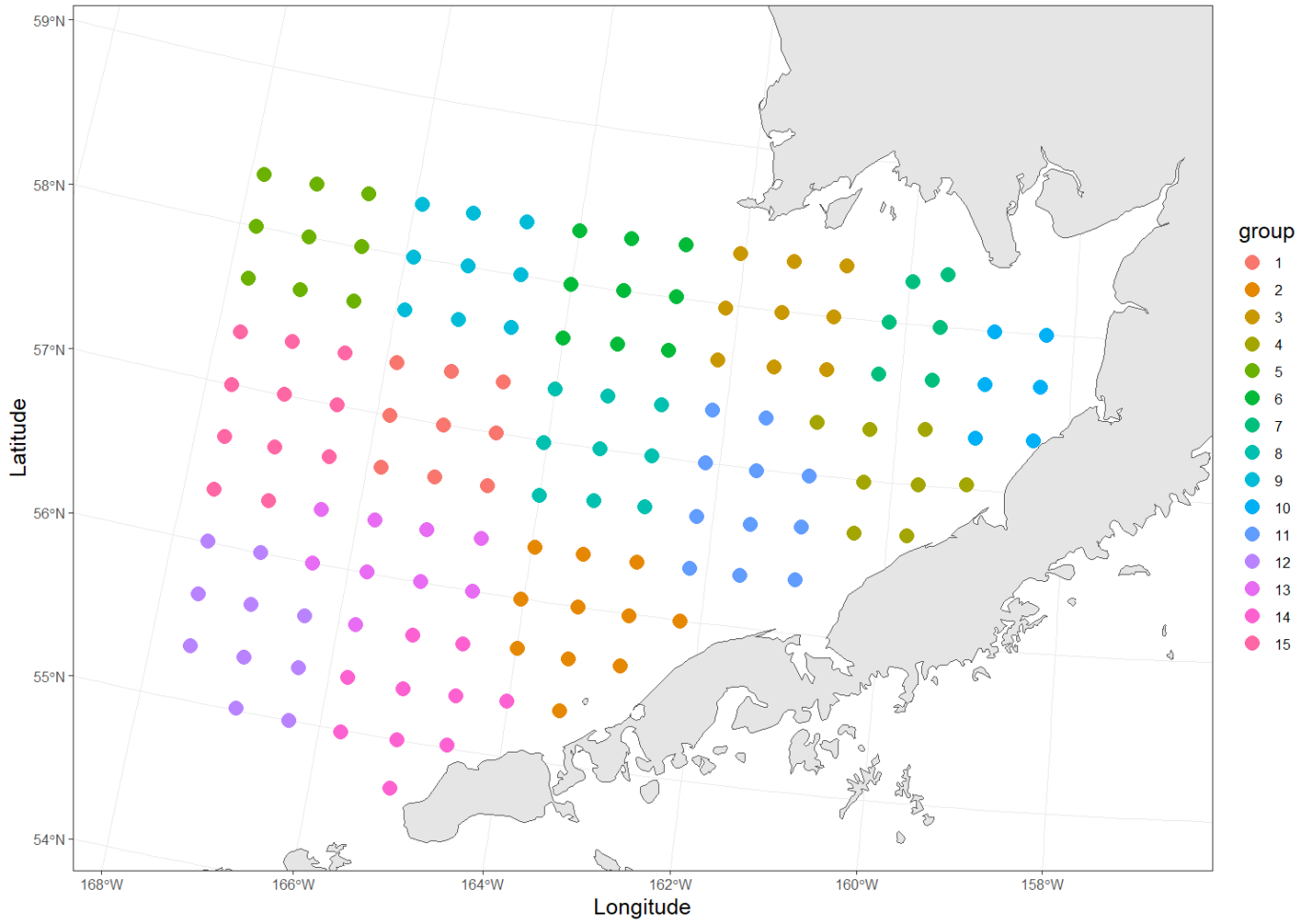


Sample variogram analysis to evaluate residual spatial autocorrelation from Tweedie GLM

- Evaluates variance among observations at varying distances (semivariance), where lower values indicate higher correlation
- More complex models will better explain residual variation and result in semivariance quickly plateauing

Suggests that residual spatial autocorrelation begin to diminish at ~75 km but does not plateau

Introducing the GLMM



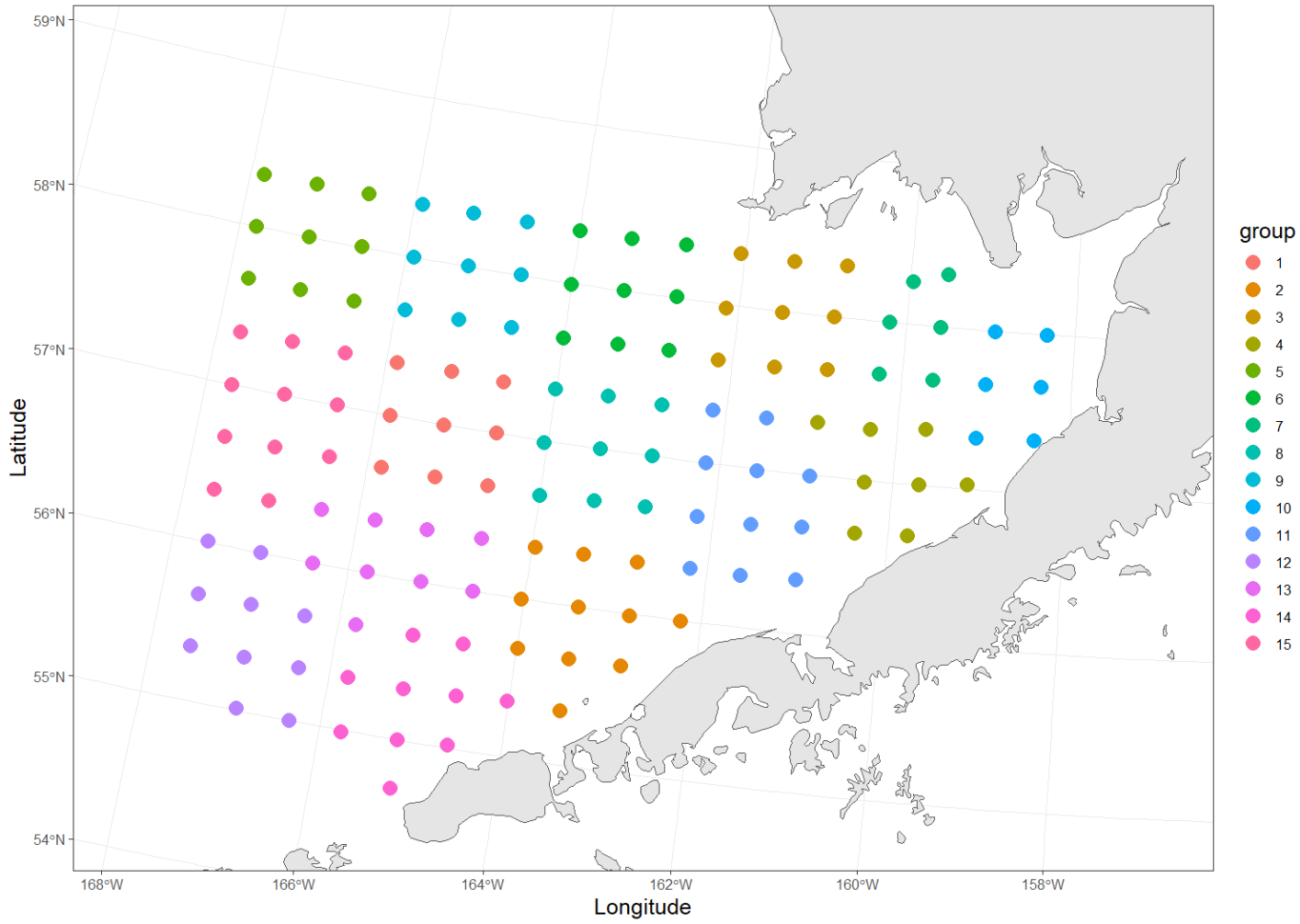
A simple option is to add spatial structure for groups of observations

Model group-level deviations from the overall mean using **random intercept** α_g

With a random intercept, our model becomes a GLMM

$$\text{CPUE}_s \sim \text{Tweedie}(\mu_s, \phi, p)$$
$$\log(\mu_s) = \beta_0 + \alpha_g$$
$$\alpha_g \sim N(0, \sigma_g^2)$$

Introducing the GLMM



What is a random intercept?

- A group-level "correction" or "modification" of the population-level intercept β_0
- Random intercepts capture within, but not between, group correlations
- Clunky but useful approach to account for residual spatial autocorrelation

$$\mathrm{CPUE}_s \sim \mathrm{Tweedie}(\mu_s, \phi, p)$$

$$\log(\mu_s) = \beta_0 + \alpha_g$$

$$\alpha_g \sim N(0, \sigma_g^2)$$

Maximum marginal likelihood estimation for GLMM

$$\mathrm{CPUE}_s \sim \mathrm{Tweedie}(\mu_s, \phi, p)$$

$$\log(\mu_s) = \beta_0 + \alpha_g$$

$$\alpha_g \sim N(0, \sigma_g^2)$$

Evaluating the likelihood for GLMMs requires computing an integral over random effects

- The general structure of this integral is

$$\mathcal{L}(\theta; Y) = \int P(Y|\theta, \epsilon) P(\epsilon|\theta) d\epsilon$$

θ - a vector of parameters (e.g.,$\beta_0, \phi, p, \sigma_g^2$)

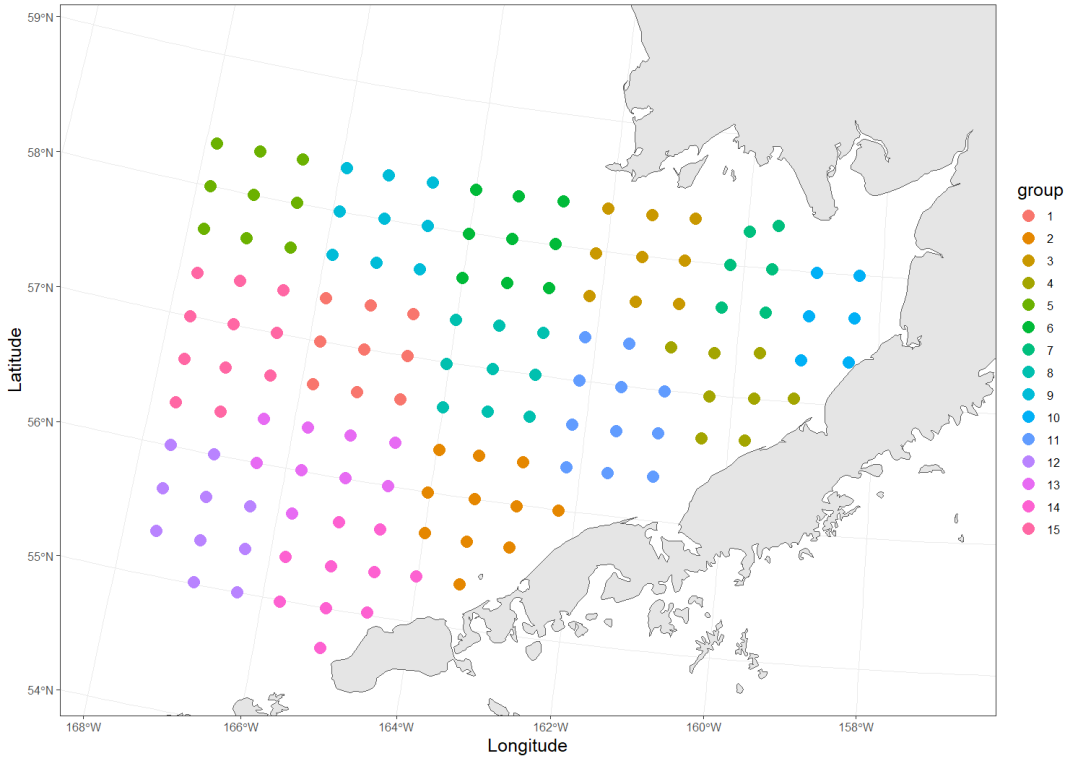
Y – observed data

ϵ – random effects (e.g., α)

Template Model Builder (TMB), the C++ software that *sdmTMB/tinyVAST* use to estimate models, evaluates this integral efficiently via the **Laplace approximation**

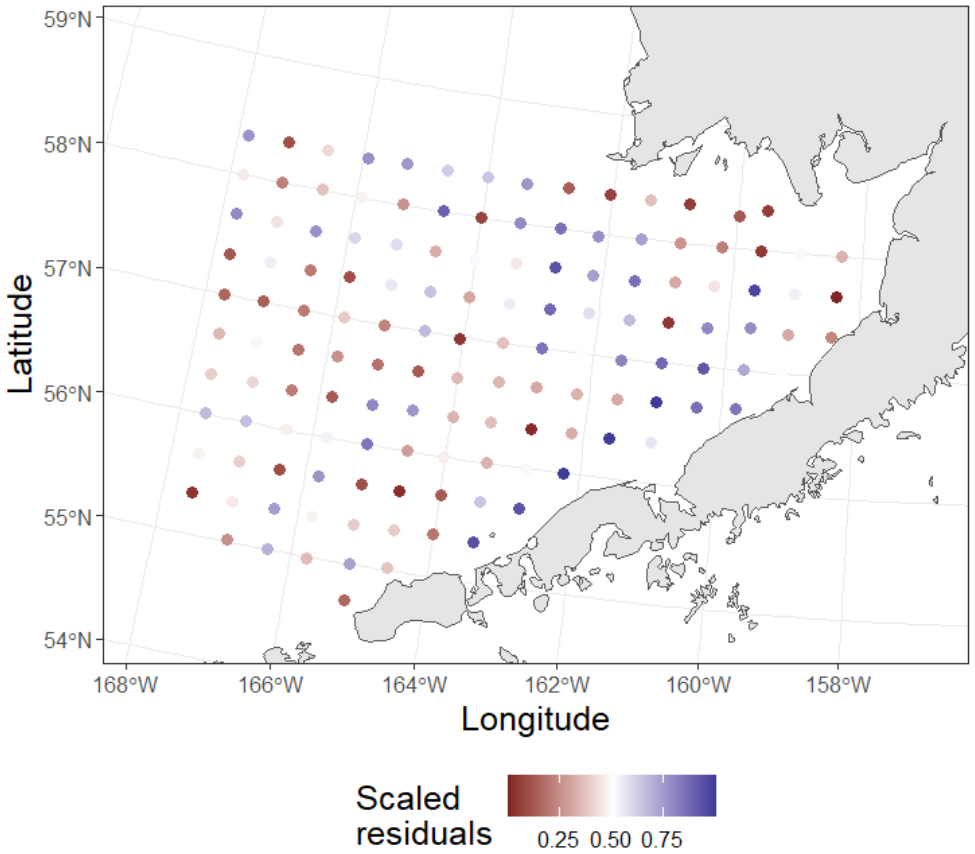
- TMB uses **automatic differentiation** to calculate gradients of the log-likelihood surface
- An optimizer uses these gradients to find the parameters that maximize the marginal log-likelihood $\mathcal{L}(\theta; Y)$

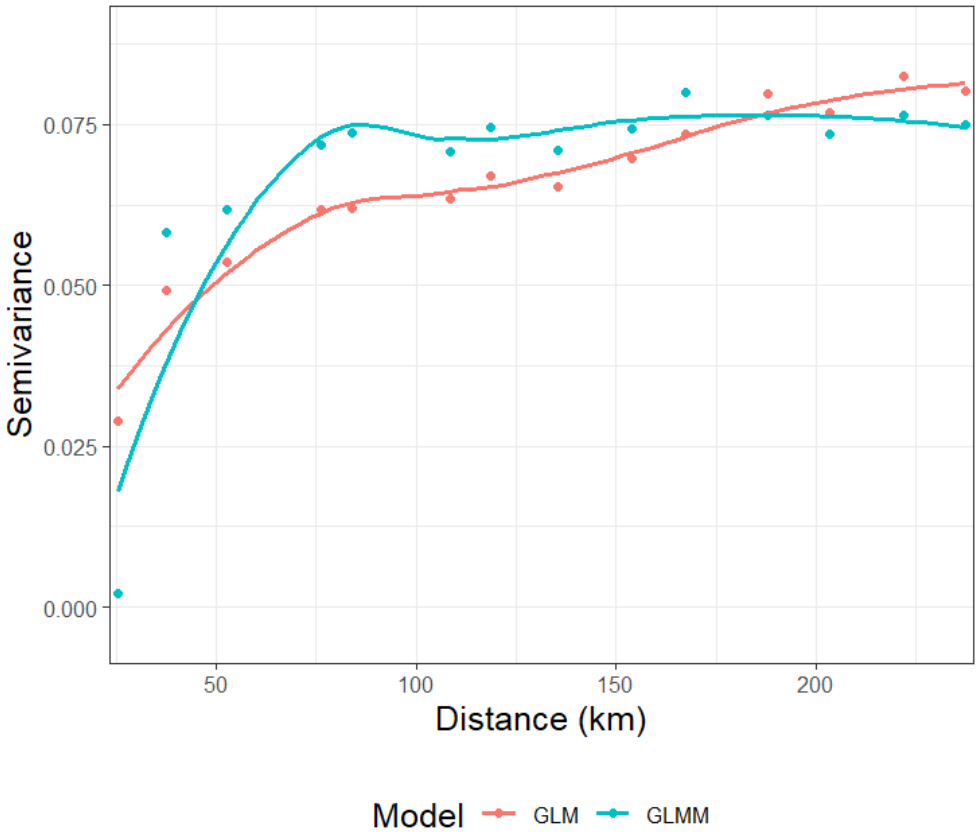
Fitting simple GLMMs in *sdmTMB*




```
> ## fit the GLMM
> mod2 <- sdmTMB(cpue ~ 1 + (1|group),
+                family = sdmTMB::tweedie(),
+                spatial = "off",
+                data = rkc_cpue_2010_df)
> mod2
Model fit by ML ['sdmTMB']
Formula: cpue ~ 1 + (1 | group)
Mesh: NULL (isotropic covariance)
Data: rkc_cpue_2010_df
Family: tweedie(link = 'log')

Random intercepts and/or slopes:

Conditional model:
     Groups        Name    Variance    Std.Dev.
      group (Intercept)        7.87        2.81

Conditional model:
            coef.est coef.se
(Intercept)      1.8    1.04

Dispersion parameter: 31.13
Tweedie p: 1.38
ML criterion at convergence: 306.906

See ?tidy.sdmTMB to extract these values as a data frame.
```

Evaluating GLMM residual spatial autocorrelation





Inclusion of random intercept reduces residual spatial autocorrelation relative to GLM

- More evenly distributed spatial residuals
- Residuals are less well correlated at shorter distances, less obvious spatial structure on map

GLM vs. GLMM predictions

The GLMM with improved model residual diagnostics has a substantially lower mean CPUE estimate:

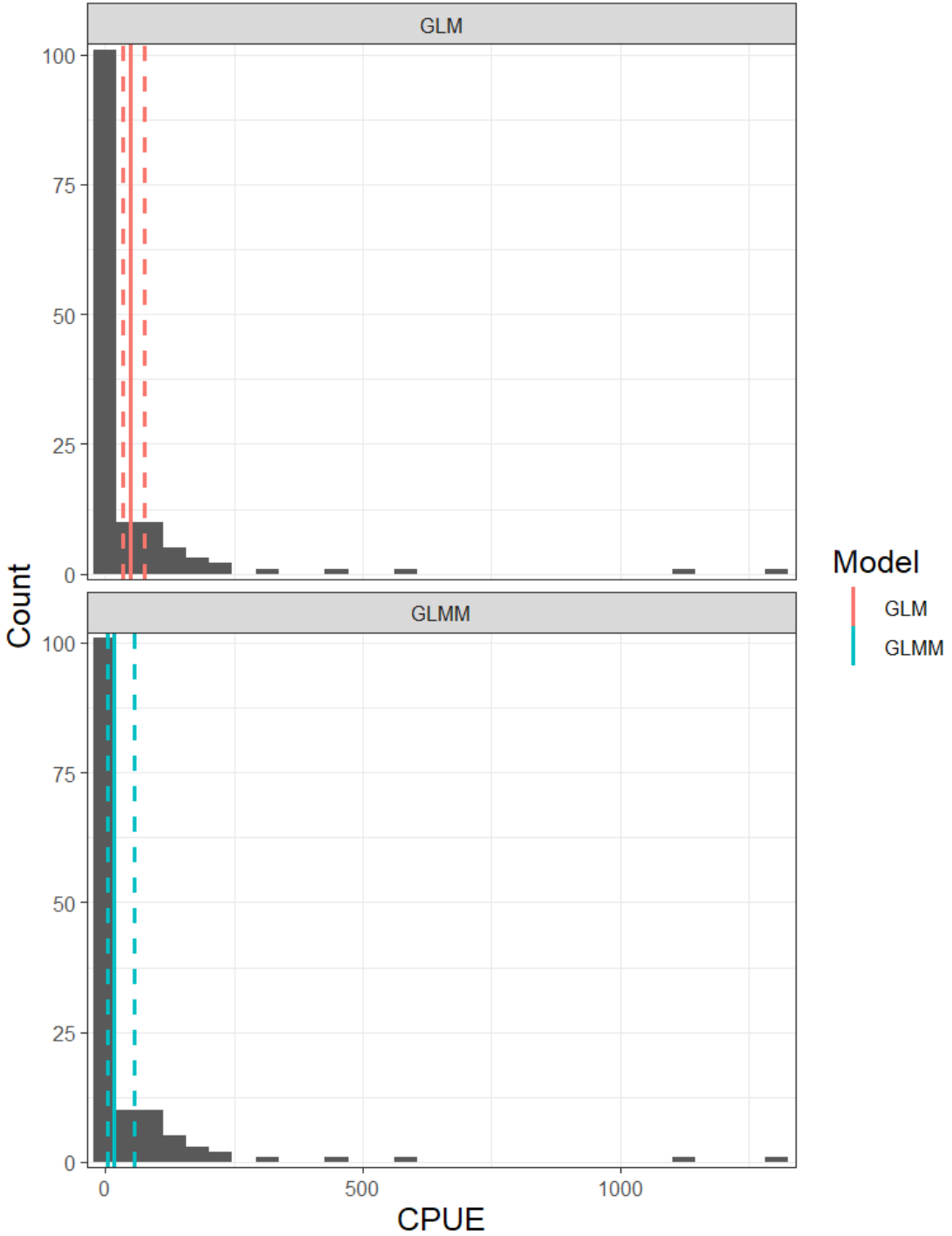
- **GLM** estimated mean = **50.1** (95% CI: 33.2 - 75.8)
- **GLMM** estimated mean = **17.0** (95% CI: 5.1 - 57.2)

The hacky random intercept improved residual diagnostics but the approach is limited:

1. The GLMM model assumes between-group residual independence despite spatially continuous correlations among observations

2. We have no way to interpolate CPUE with uncertainty at unmeasured locations

How can we do better? **Spatial GLMM**



Intuition for Gaussian fields and spatial GLMM

Generically, suppose we collect samples of some variable over a continuous spatial domain

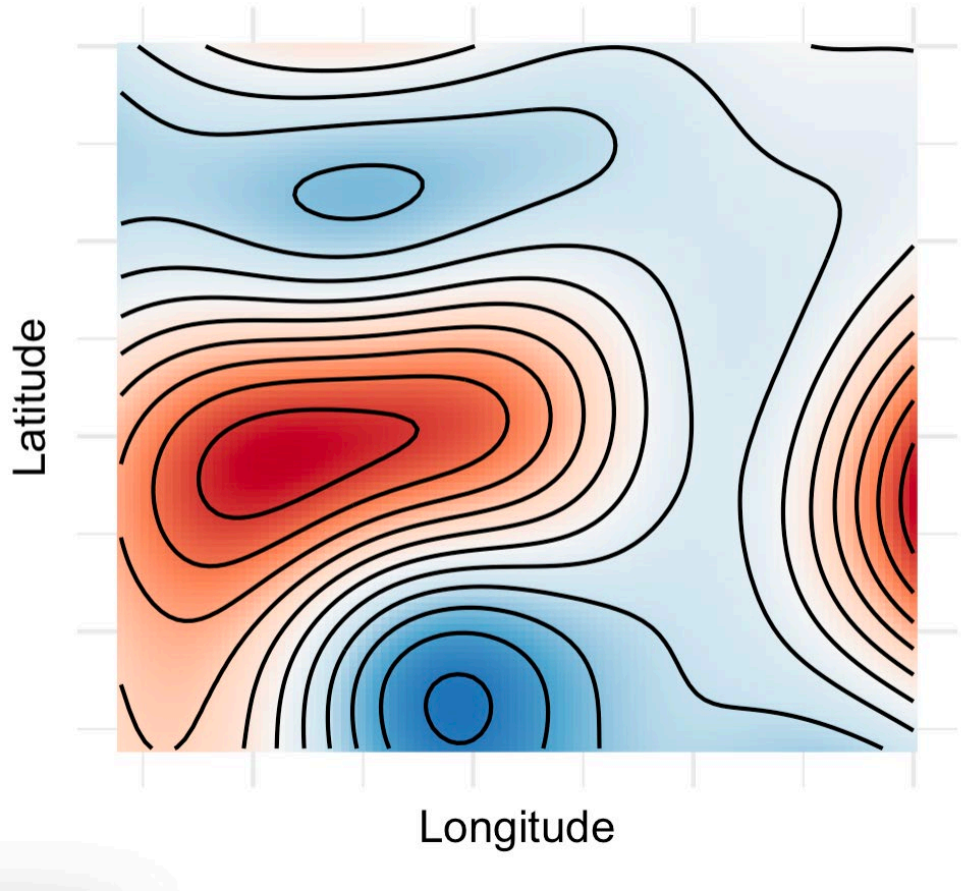
These samples are draws from a underlying *stochastic process*

- If we repeat sampling under the exact same conditions, we will end up with different realizations of the same process

If we assume observations are Gaussian, then we have a Gaussian Field

In our spatial GLMM

- We model unexplained (by fixed effects) variability in CPUE using a residual term, ω
- ω is modeled as a GF defined by mean and covariance functions – no longer assumed iid



Generic example of a 2D Gaussian Field

Spatial GLMMs

Modeling the covariance matrix $\boldsymbol{\Sigma}_\omega$ allows for making predictions with uncertainty at unmeasured locations – *kriging*

$$\mathrm{CPUE}_s \sim \mathrm{Tweedie}(\mu_s, \phi, p)$$

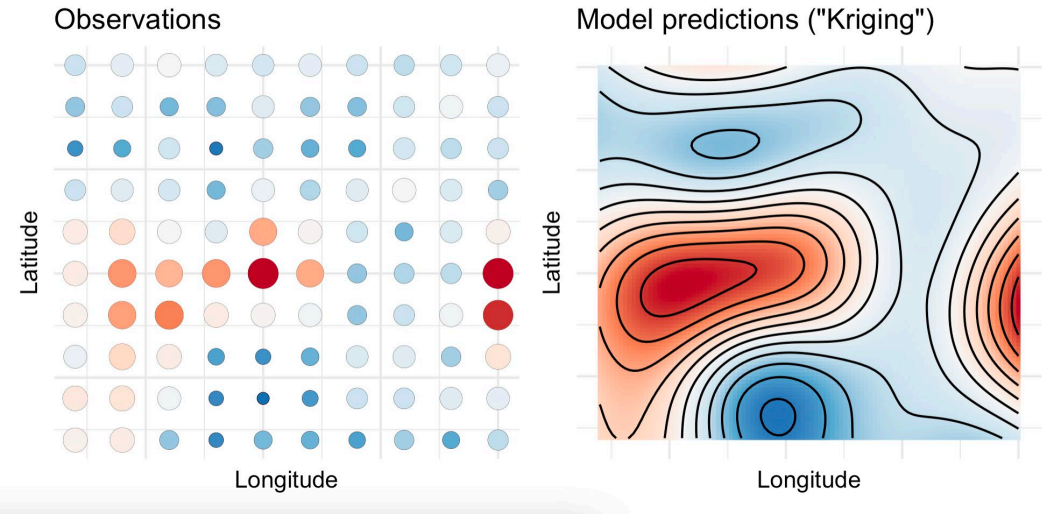
$$\log(\mu_s) = \beta_0 + \omega_s$$

$$\boldsymbol{\omega} \sim \mathrm{GF}(\mathbf{0}, \boldsymbol{\Sigma}_\omega)$$

The elements of covariance matrix $\boldsymbol{\Sigma}_\omega$ are covariances between all $\boldsymbol{\omega}$, too many to estimate

Instead, we calculate covariances based on

1. Euclidean distances between observations
2. A function representing how spatial covariance decays with distance, the **Matérn covariance function**



Generic example of kriging interpolation

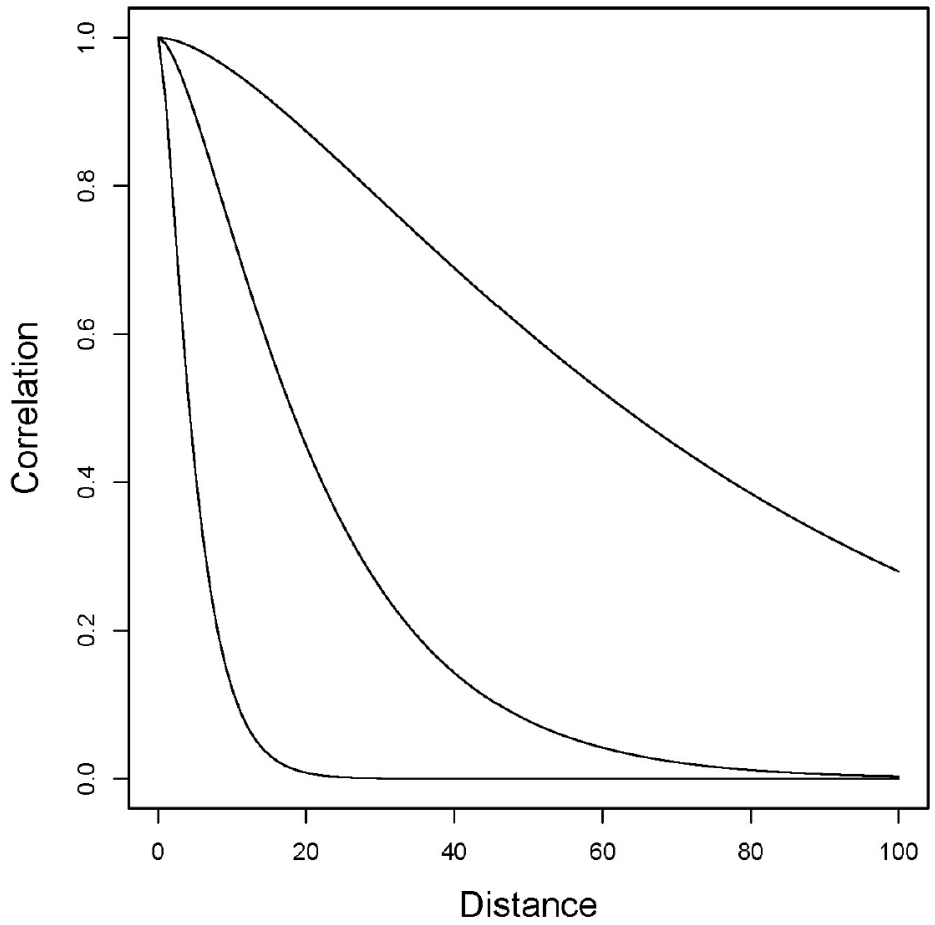
Matérn covariance

In *sdmTMB*, calculating Matérn covariance requires estimating two parameters:

1. κ – a scale parameter where the **spatial range** = $\sqrt{8}/\kappa$
2. σ_m^2 - the marginal variance aka the 'magnitude of random field "wiggles" '

Given these parameters and the Euclidean distance between observations we can fill-in the spatial covariances of $\boldsymbol{\Sigma}_\omega$

An assumption of Matérn covariance is **isotropy** – that correlation decays equally in all directions



Three Matérn correlation functions for different k and $\nu = 1$ (Zuur et al. 2017)

Computational shortcuts: GF ⟶ GMRF

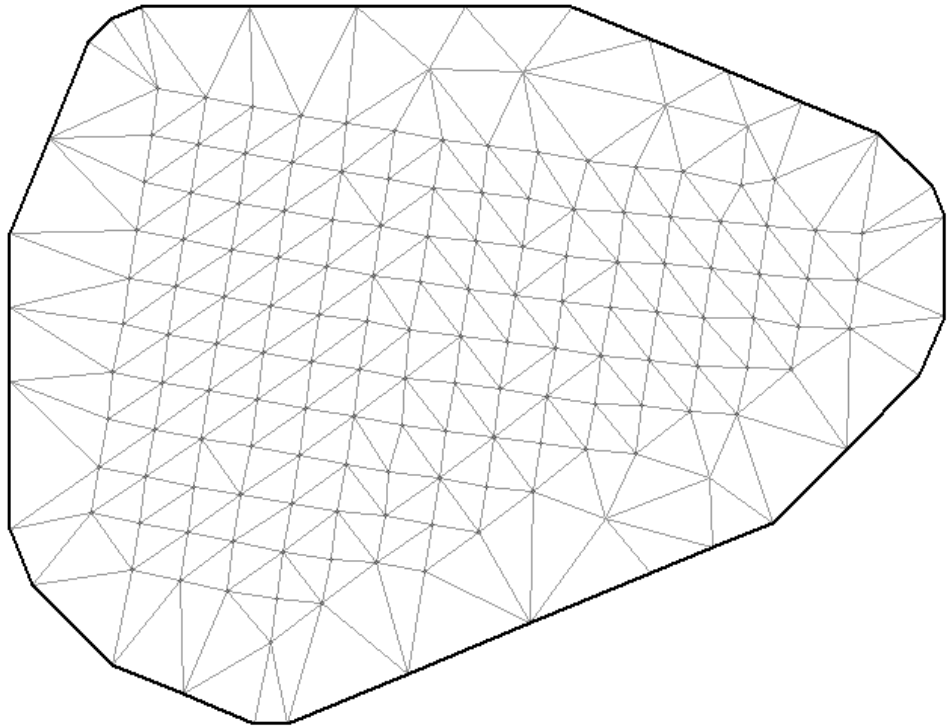
Working with dense covariance matrices like $\boldsymbol{\Sigma}_{\omega}$ is computationally expensive

To improve efficiency, we approximate the GF as a **Gaussian Markov Random Field (GMRF)**

- The precision matrix Q for the GMRF is sparse and easier to work with than the dense covariance

This approximation is done using the **stochastic partial differential equation (SPDE)** approach

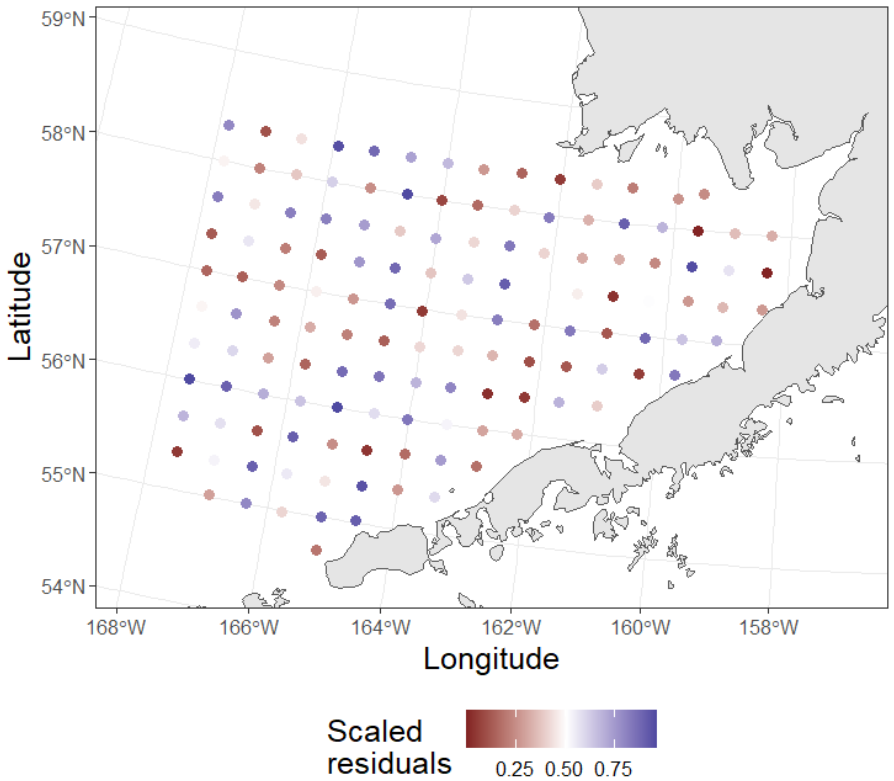
The approximation is performed using a finite element mesh (*"the mesh")* over the spatial domain

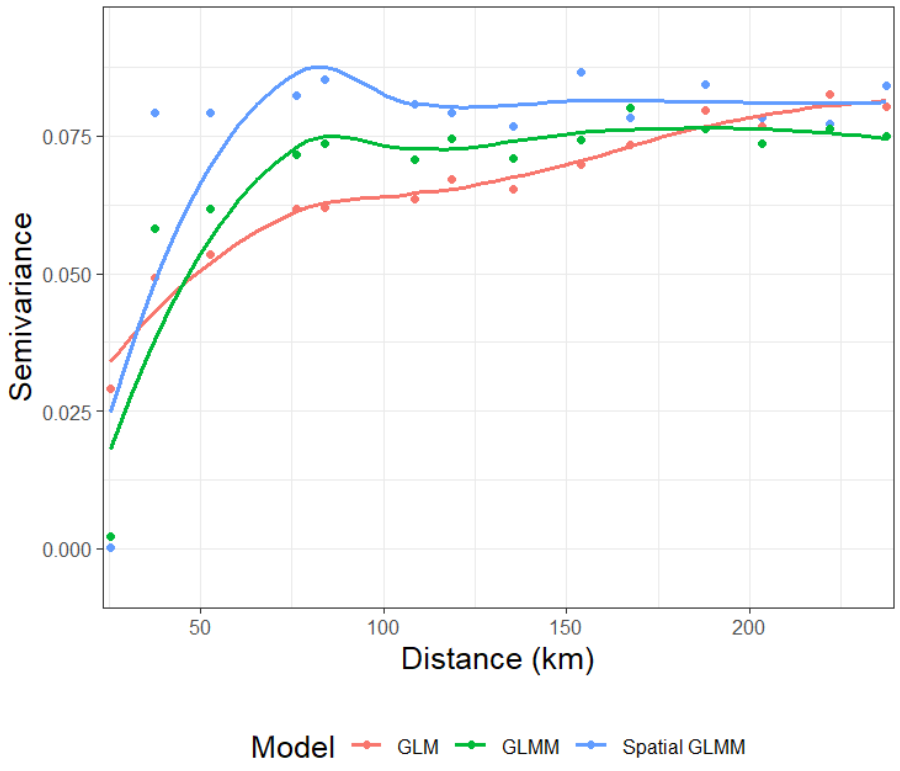


A finite element mesh defined over Bristol Bay

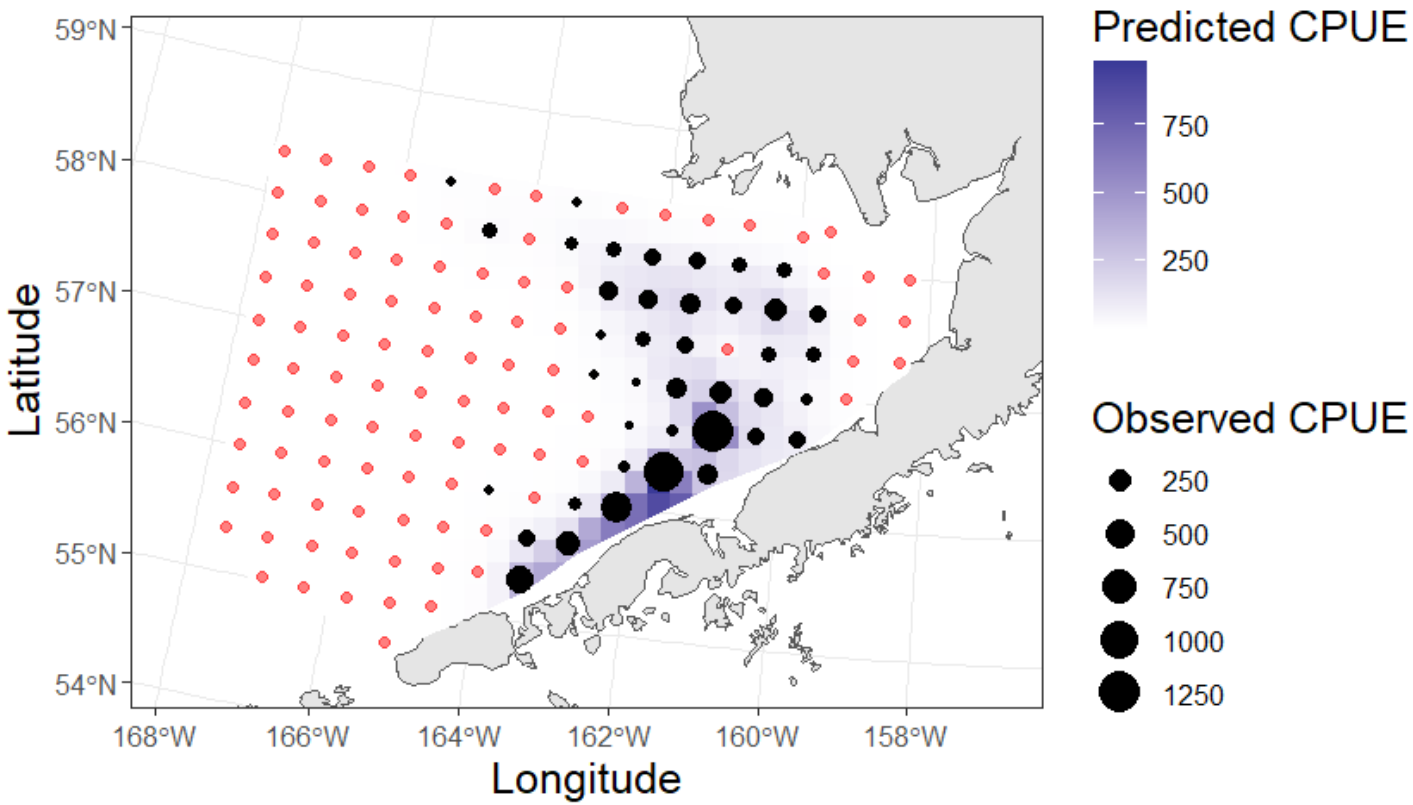
Fitting the spatial GLMM

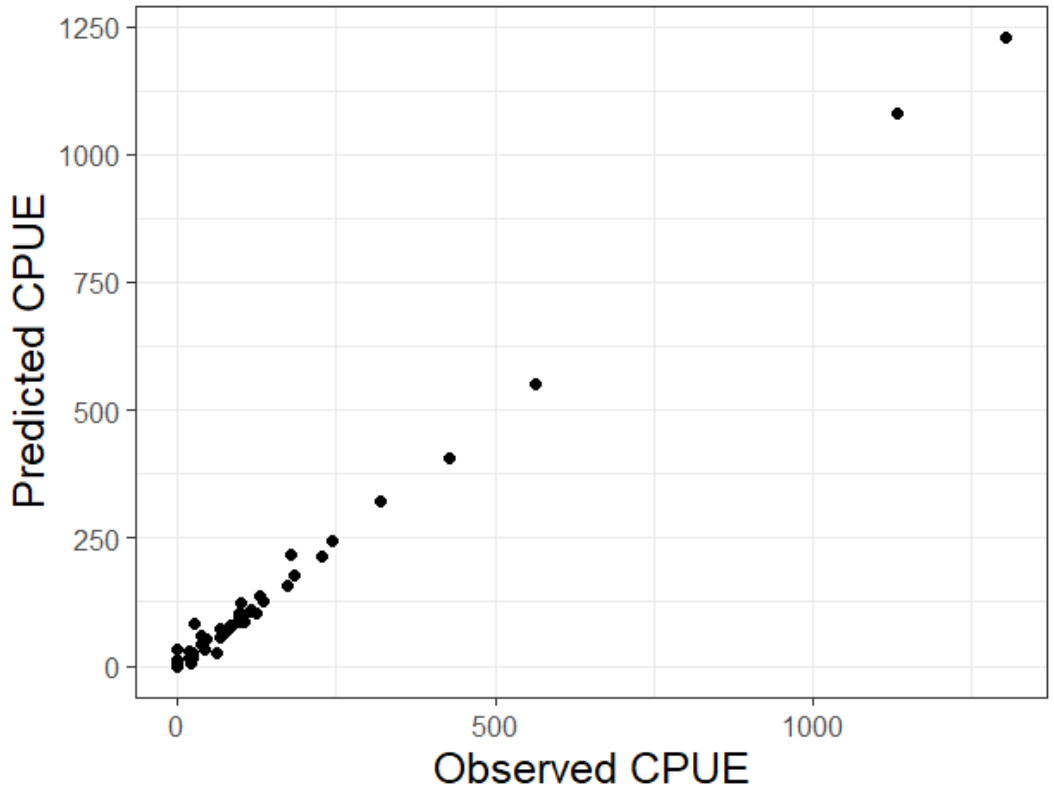
```
> # Spatial GLMM-------------------------------------
> rkc_mesh <- make_mesh(data = rkc_cpue_2010_df,
+                       xy_cols = c("longitude", "latitude"),
+                       type = "cutoff", cutoff = 10)
> mod3 <- sdmTMB(cpue ~ 1,
+                family = sdmTMB::tweedie(),
+                spatial = "on",
+                mesh = rkc_mesh,
+                data = rkc_cpue_2010_df)
```





Fitting the spatial GLMM





Expanding into space and time: IID spatiotemporal random fields

The GMRF estimates latent spatial variability in a residual term

Spatial processes evolve through time

We can model spatiotemporal random fields as temporally independent ("iid") or spatiotemporally correlated (AR1/random walk)

- Spatiotemporal correlation structure is required for spatiotemporal interpolation and projection

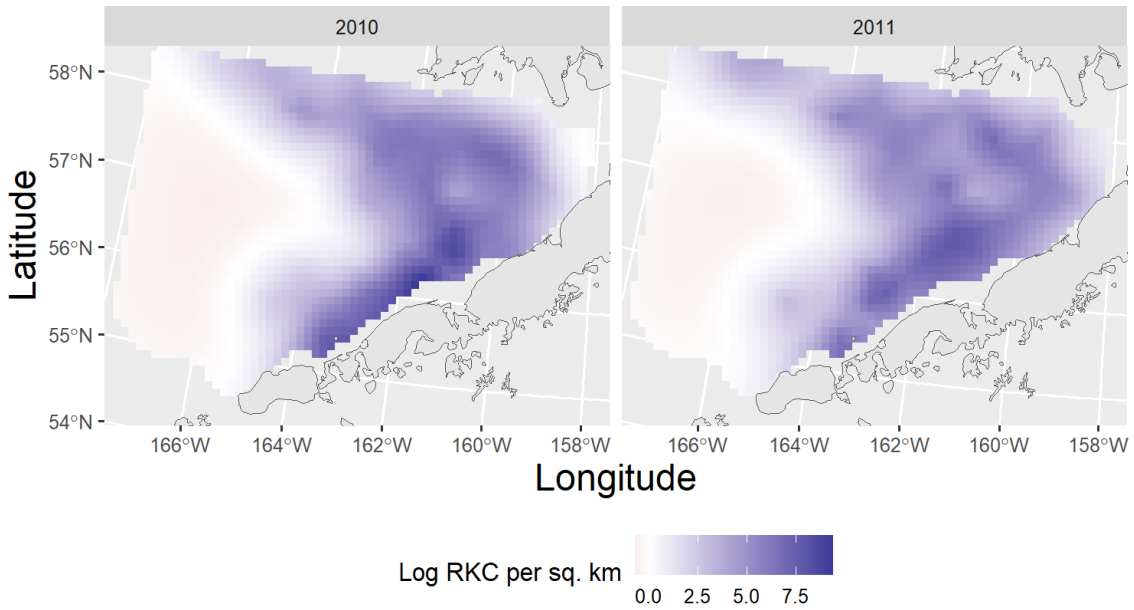
We will explore these correlation structures in detail in the following demonstration and labs

$$\mathrm{CPUE}_{s,t} \sim \mathrm{Tweedie}(\mu_{s,t}, \phi, p)$$

$$\log(\mu_{s,t}) = \beta_0 + \omega_s + \epsilon_{s,t}$$

$$\boldsymbol{\omega} \sim \mathrm{GMRF}(\mathbf{0}, \boldsymbol{\Sigma}_\omega)$$

$$\boldsymbol{\epsilon}_t \sim \mathrm{GMRF}(\mathbf{0}, \boldsymbol{\Sigma}_\epsilon)$$



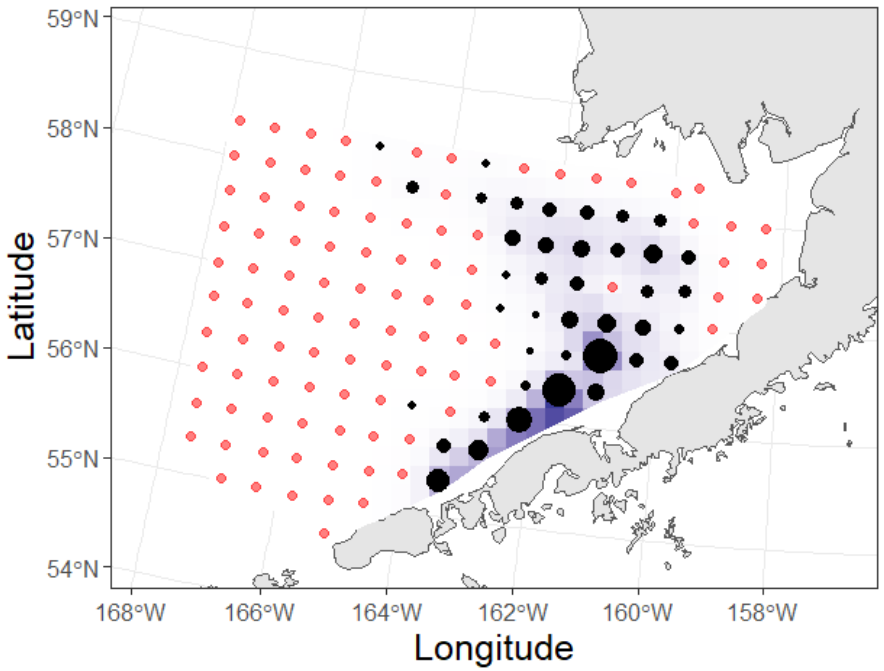
Recap

GLMs are useful for modeling data drawn from diverse probability distributions

- Fitted with MLE that assumes residual independence
- DHARMa residuals/sample variograms

Simple GLMM with random intercepts is useful but limited

- Spatial autocorrelation is continuous, not discrete + no capacity for interpolation/projection
- These and more complex models with random effects/fields fitted using *marginal maximum likelihood estimation*

Spatial and spatiotemporal GLMMs

- Involve estimating Matern covariance; approximating the GF as a GMRF via SPDE method
- Resolves residual spatial autocorrelation to improve inference
- Relies on spatial covariance to skillfully predict